

A local Model Context Protocol server that lets Claude Code, Codex, and Claude Desktop answer natural-language questions about a ~62,000-ticket support corpus — with hybrid retrieval, verbatim citations, and a grounded draft-reply prompt assembled entirely on-device.

DESIGN — WHAT'S BUILT AND HOW IT ANSWERS QUESTIONS

Query → BM25 (top 50) ♦ vector kNN over e5-small embeddings (top 50) → Reciprocal Rank Fusion → top 10 previews + ticket://{id} citations → client LLM reasons over snippets and (optionally) drafts a reply.

One Python process exposes the dataset to any MCP client over stdio. Rows, BM25 index, and 384-dim vectors live together in LanceDB; aggregations run zero-copy through Polars over Arrow. Stable 12-char ticket ids are derived from sha1(revision || row_index) so citations survive across sessions and machines.

Surface	Kind	Purpose
search_tickets	Tool	Hybrid BM25 + vector + RRF, with filters (queue, priority, language, type, tags)
get_ticket(id)	Tool	Returns one ticket, verbatim, every column
aggregate_tickets	Tool	Counts grouped by queue / priority / language / type / tags
create / update / delete_ticket	Tools	Local CRUD; re-embeds on write so changes are immediately searchable
server_info	Tool	Revision pin, embedding model id, row count, license
ticket://{id}, schema://{tickets}	Resources	Citation handles and column-value catalog
draft_reply	Prompt	Target ticket + up to 5 grounding neighbours (cosine ≥ 0.70) + type-aware scaffold; client's LLM writes the reply

WHY — THE SOLUTION THAT FITS THIS DATASET

This dataset. Messy free text, bilingual (EN+DE), no timestamps, no ids. A churn-style tabular set reduces to text-to-SQL; this corpus forces the interesting choices — retrieval architecture, multilingual embeddings, grounded generation — and every row ships with the human support answer, giving draft_reply real prior replies to anchor on.

Hybrid retrieval. BM25 wins on literal tokens (error codes, SKUs); vector wins on paraphrase ("login broken" ≈ "can't sign in"). RRF merges by rank position — no score normalisation, no weights to tune.

LanceDB + e5-small. One embedded columnar store holds rows, BM25, and vectors; scales to 100M+ on the same API. multilingual-e5-small (384-dim) is free, local, multilingual.

Deliberately avoided. Cross-encoder reranking (small win when an LLM re-reads snippets anyway); HyDE / query rewriting (upstream LLM already phrases well); chunking (tickets are 100–2200 chars — embed whole); remote HTTP + OAuth (local stdio is the assignment); server-side LLM calls (no API-key dependency).

RISKS — WHAT GOES WRONG, AND HOW IT'S PREVENTED

Risk	Mitigation
Hallucinated tickets	Read endpoints return dataset bytes verbatim; server never calls an LLM; draft_reply only assembles context + scaffold.
Prompt injection via ticket bodies	Content wrapped in <ticket> tags; tool descriptions label inner text as data; draft_reply rejects injection-shaped inputs.
Dataset drift	HF revision pin embedded in every ticket://{ URI and the embedding cache key; re-index triggers on revision change.
Confident answers from weak matches	Return relevance scores + snippets, not just ids; fail loud on unsupported filters (e.g. dates — no timestamp column); precision@5 in prod.
License (CC-BY-NC-4.0)	Surfaced in server_info and README; production swaps to a licensed or owned corpus.
Cross-lingual quality (e.g. Hebrew)	Documented limitation; one-line upgrade path to bge-m3.
First-run latency (~2 min embed pass)	Progress via MCP notifications/progress; cached on disk by revision + model id; sub-second on subsequent starts.

PRODUCTION DELTAS — WHAT CHANGES FOR REAL

- **Live source & incremental index.** Real ticketing system instead of the pinned HF snapshot; periodic delta-ingest, not full re-embed.
- **Remote transport + OAuth 2.1.** Streamable HTTP, audience-bound tokens, multi-tenant isolation (tenant id partitions the vector index).
- **Reranker on by default.** bge-reranker-base once retrieval feeds real workflows.
- **Upgrade embeddings to bge-m3** on a small GPU for stronger multilingual (incl. Hebrew) coverage.
- **PII redaction** at ingest (email, phone, IBAN, names) — deferred for the public demo set.
- **Eval harness.** 50–200 golden queries with expected ids; precision@5 tracked per deploy.
- **Observability.** Structured logs, traces, p95 latency & retrieval-quality dashboards; per-tenant rate limits.
- **Supply chain.** Reproducible builds, signed images, CVE gating, model-artifact pinning.

BUSINESS IMPACT — HOW THIS CREATES VALUE AT CHEQ

The pattern — hybrid retrieval over a text-rich corpus, grounded generation with verbatim citations, exposed through MCP — is the integration layer for putting internal data under one trustworthy contract across every downstream AI workflow.

- **Fraud-incident triage** direct fit. Same architecture, different corpus: BM25 catches exact indicators (IPs, hashes, domains, error codes); vectors catch behavioural patterns described in analyst prose. Analyst chats "find prior incidents matching this pattern from the same IP range", every cited incident a real record.
- **Support & CSM copilot.** Pointed at CHEQ's real ticket / Slack / Salesforce history: "find prior tickets where a customer hit error X on integration Y", "draft a reply matching how we resolved it last time" — every claim traceable.
- **Threat-intel & rules knowledge base.** Detection rules, intel notes, fraud playbooks searchable under the same grounding contract — instead of N one-off RAG bolt-ons.
- **Why MCP.** Build the server once, plug it into Claude Code, Codex, Cursor, internal agents. MCP makes grounded retrieval composable across the company; the alternative is each team re-solving hallucination, citation, and access control alone.